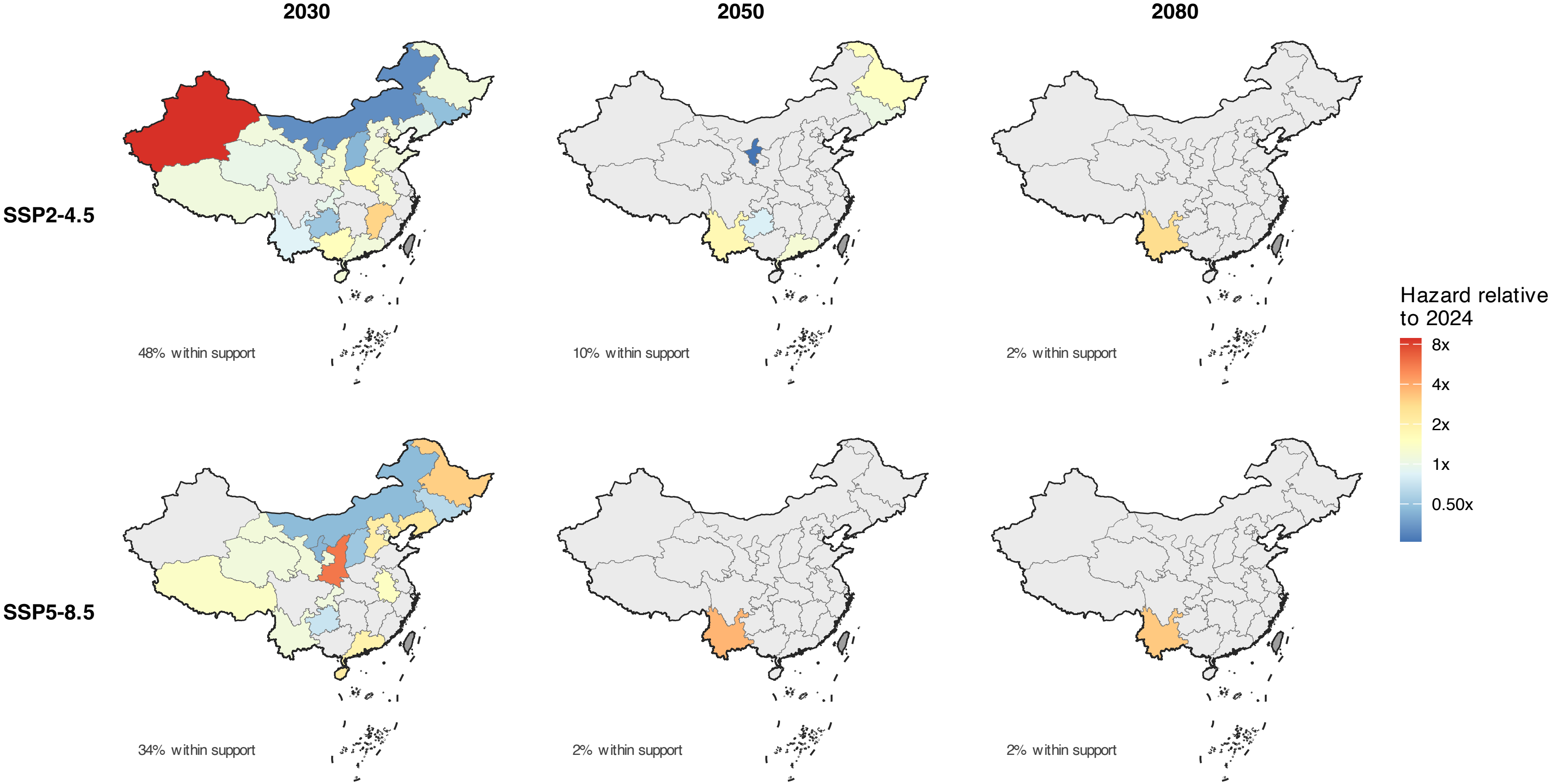
Machine-learning projection of new-record generation hazard

XGBoost on the same three predictors and identical future covariates; only rows inside the fitted covariate support



Latent generation hazard relative to the 2024 baseline (log2 colour scale); province-by-year shocks set to expectation and reporting completeness set to 1. Base map GS(2019)1822. Pale grey: inside the analysis scope but no cell falls within the fitted covariate range for that scenario and horizon. Mid grey: outside the analysis scope (Taiwan, Hong Kong, Macao and the jointly administered area, none of which have provincial effort coverage). Gradient boosting returns boundary leaf values outside the training range, so its surface flattens rather than extrapolating.